

Moonmist

Tags:

Time Limit: 10 s Memory Limit: 64 MB
Submission: 102 AC: 21 Score: 99.18

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Codes

AI Code Tips

Description

An Unidentified Flying Object (Commonly abbreviated as UFO) is the popular term for any aerial phenomenon whose cause cannot be easily or immediately identified. We always believe UFO is the vehicle of aliens. But there is an interrogation about why UFO always likes a circular saucer? There must be a reason. Actually our scientists are developing a new traffic system "Moonmist". It is distinguished from the traditional traffic. We use circular saucers in this new traffic system and the saucers moves extremely fast. When our scientists did their test, they found that traffic accident was too hard to be avoided because of the high speed of the advanced saucer. They need us to develop a system that can tell them the nearest saucer. The distance between two saucers is defined as the shortest distance between any two points in different saucers.

Input

The first line consists of an integer T , indicating the number of test cases.

The first line of each case consists of an integer N , indicating the number of saucers. Each saucer is represented on a single line, consisting of three integers X, Y, R , indicating the coordinate and the radius. You can assume that the distance between any two saucers will never be zero.

Output

For each test case, please output a floating number with six fractional numbers, indicating the shortest distance.

Constraints

 $0 < T \leq 10$ $2 \leq N \leq 50000$ $0 \leq X, Y, R \leq 100000$

Samples

input

Copy

```
1
2
0 0 1
10 10 1
```

output

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12.142136

Source

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[HZNUOJ](#) is based on [HUSTOJ](#)

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